

THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

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Anonymous submission

ABSTRACT

We establish how physical scale and population size shape the emergence of complex behavior in open-ended ecological environments. In our setting, agents receive no explicit rewards, supervised targets, or hand-designed objectives; instead, they evolve through reproduction, mutation, and selection. As agents act, they continuously reshape both the environment and the population around them, producing an ongoing dynamic ecology rather than a fixed optimization problem. Our objective is therefore not to train a single high-performing policy, but to characterize how behaviors arise and stabilize across large populations under natural competition and environmental pressure. We combine modern accelerators with a new multi-agent simulator to scale ecological worlds far beyond prior attempts, reaching populations of more than 60,000 agents, each carrying its own evolved neural network policy. Across these worlds, we identify robust emergent behaviors including long-range resource extraction, vision-guided foraging, and predation. Our experiments show that sensing modalities and environmental scale decisively affect which behaviors emerge: several behaviors appear only at sufficiently large scales, and larger environments make successful behaviors more stable and reproducible across seeds. Although evolutionary computation has a long history, these scaling results demonstrate that modern hardware now makes ecology a practical and powerful instrument for machine learning research, opening a concrete path toward studying intelligence through large evolving populations. Experimental code will be released upon acceptance.

1 INTRODUCTION

Since Darwin, the evolutionary emergence of biodiversity and complex behavior has anchored research across evolutionary theory (Darwin, 1859; Simpson, 1944), ecology (MacArthur & Wilson, 2001), and genetics (Dobzhansky, 1937). These fields have developed a rich account of emergent dynamics, from theories of speciation and adaptive radiation (Coyne & Orr, 2004; Schluter, 2000) to the mathematical foundations of population genetics (Fisher, 1930; Wright, 1931).

Decades of mathematical modeling, laboratory experimentation, and field work have revealed many mechanisms that drive evolution in nature. Yet natural ecosystems remain intrinsically difficult experimental systems: as they grow in scale and complexity, they become harder to control, instrument, and replicate. Even when intervention is possible, controlled experimentation in large natural settings can damage or displace wild populations.

We address these barriers by studying the open-ended evolution of neural network policies in large-scale ecological simulations. Agents receive no specified objective or reward signal; they must instead collect resources to survive and reproduce under the dynamics of the environment. Policy change occurs only through mutation when parents pass policies to their children. The rapid expansion of available compute now makes it possible to study these digital organisms at ecological scales that were previously out of reach. This work uses that opportunity to directly examine how environmental scale and population size govern the emergence of behavior.

To do so, we introduce a JAX-based simulation environment that supports rapid evaluation of large grid worlds. Our largest experiments contain more than 60,000 individual agents and more than 1,000,000 grid cells. Within these worlds, agents must navigate to locate food and other resources,

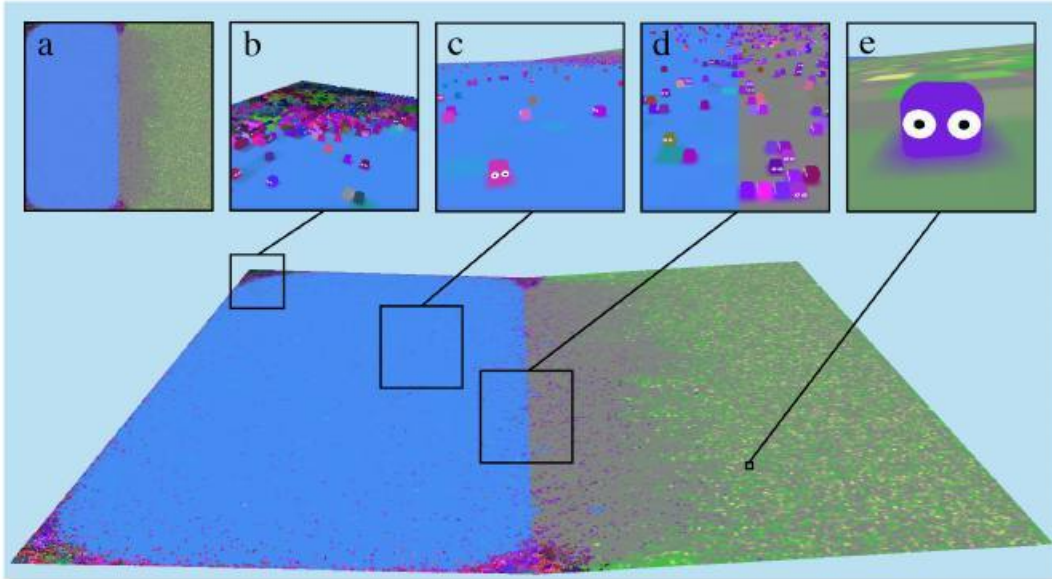


Figure 1: A 3D visualization of a 512×512 map in our environment. Inset (a) gives a top-down view of the full map, (b) highlights agents clustered in a corner, (c) shows agents in water, (d) shows agents traveling between land and water, and (e) shows a single agent gathering resources.

and agents that accumulate sufficient resources reproduce by creating mutated copies of themselves. Figure 1 provides a 3D visualization of the environment.

Our aim is not only to simulate ecological phenomena, but to position ecology itself as a mechanism for producing machine intelligence. Recent AI systems show that scale does not merely improve performance; it can transform capabilities (Wei et al. 2022a). Yet embodied and ecological learning studies have often remained confined to small environments and small populations. Here, we scale not a single model but an entire evolving ecosystem. Just as reasoning capabilities can appear in individual models only beyond particular scale thresholds (Wei et al. 2022b), we investigate which strategies become available to populations only when the environment is large and rich enough to support them.

Our experiments identify multiple cases in which sensor configurations and environmental scale determine the behaviors that emerge and the long-term ecology that follows. Agents with simple compass sensors reliably evolve long-range resource-gathering expeditions, while agents without compasses usually fail to maintain the same behavior. Vision sensors produce more effective foraging and attacking behavior than blind variants. In both settings, the relevant behaviors appear more reliably and stabilize more strongly at larger scales. Together, these results establish large-scale ecological simulation as a concrete experimental platform for studying how evolution can produce rich behavior without explicit task objectives.

2 RELATED WORK

The emergence of life has motivated computational research since the field’s inception. Conway’s Game of Life (Games 1970) showed how simple local rules can generate population-level dynamics. Tierra (Ray 1992) and Avida (Brown) advanced this agenda by simulating self-replicating programs that evolve through mutation, while Reynolds (1987) demonstrated that local motion rules can produce flocking and schooling. These systems established the central premise that complex collective behavior can arise from simple distributed mechanisms.

Evolutionary methods have also produced virtual-agent morphology (Sims 1994a,b; Ha 2018; Gupta et al. 2021; Auerbach et al. 2014; Auerbach & Bongard 2012; Jeon et al. 2025; Auerbach & Bongard 2014; Szerlip & Stanley 2013) and sensing (Pratt et al. 2022; Mugan & MacIver 2020; Tiwary et al. 2025), and neuroevolution has optimized neural network architectures directly (Stan-

ley & Miikkulainen, 2002, 2004, Stanley et al., 2009). Our work takes a different stance: rather than using an objective or reward signal to make agents adapt better to a task, we use ecological pressure itself as the driver and ask which behaviors emerge, how environmental complexity shapes them, and when scale makes them reliable.

Evolution has likewise shaped the design of optimization algorithms. Genetic algorithms (Holland, 1992; Jong, 1975) and evolutionary strategies (Wierstra et al., 2014; Hansen & Ostermeier, 2001; Rechenberg, 1973; Schwefel, 1995; Conti et al., 2018) operationalize mutation and selection as optimization procedures. Novelty-driven methods further use evolutionary search to produce diverse solutions (Lehman & Stanley, 2011; Cully et al., 2015), and related approaches have optimized neural networks for Atari and Mujoco locomotion tasks (Salimans et al., 2017; Such et al., 2017; Mania et al., 2018). Although our agents are updated through evolutionary mechanisms, the contribution is not a new optimizer and not a stronger solver for a fixed benchmark. The contribution is an objective-free ecological setting in which behavior, population dynamics, and environmental change co-develop.

A complementary line of work uses biological inspiration to develop organisms within simulated environments. Early examples include Bedau (1992), Yaeger et al. (1994), Channon & Dampier (1998), and Tisue et al. (2004). More recent systems use cellular automata to grow organisms with emergent morphologies and behaviors (Mordvintsev et al., 2020; Robinson, 2021; Heinemann, 2021; Randazzo & Mordvintsev, 2023; Caussan, 2025). Mordatch & Abbeel (2018) and Park et al. (2023) study the emergence of language and communication between agents. Autocurricula (Leibo et al., 2019), in which populations create an increasingly challenging learning landscape for one another, have also been explored in environments with smaller populations than ours (Leibo et al., 2018; Team et al., 2021). In addition, Wang et al. (2019) and Aki et al. (2024) evolve environments designed to improve agent learning. These works underscore a point central to our paper: environmental complexity can elicit sophisticated behavior. We extend that point by showing that scale and population dynamics are themselves decisive experimental variables.

The closest prior work simulates populations of agents interacting in complex environments. Baker et al. (2019) study a Hide-and-Seek task in which a small number of hider agents learn to use tools to evade seeker agents, with both sides trained by reinforcement learning. Other systems train populations for specific tasks such as playing video games (Jaderberg et al., 2019), manipulating objects (Petrenko et al., 2021), or capturing agents (Zheng et al., 2017; Yamada et al., 2020). However, these studies typically examine emergence under fixed tasks and explicit objectives. Neural MMO (Suarez et al., 2019, 2023) moves closer to open-ended reinforcement learning by placing agents in a massively multi-agent game and optimizing survival rather than a narrow task. However, its population is capped at a fixed number of agents, with experiments up to 128 concurrent agents and without generational inheritance, which limits analysis of evolving population dynamics. Hamon et al. (2023) is closer to our setting through variable population sizes, genetically inspired evolution, and generational timelines, but its environment is substantially smaller, less feature-rich, and does not directly isolate the role of scale. Similarly, Lu et al. (2024) simulates agents evolving to gather resources, but operates at a much smaller scale, with at most 256 agents, and centers tool use through programmable robots. These prior systems motivate the need for a setting in which populations grow and shrink naturally, policies evolve without predefined objectives, agents directly interact, and the environment is large enough for collective ecological patterns to emerge. Our platform combines these ingredients and thereby exposes ecological phenomena that were inaccessible at smaller scale.

Finally, we build on recent work emphasizing simulation speed and scale through optimized libraries that increase the number of agents and interactions available in game-like environments (Koyamada et al., 2023; Matthews et al., 2024; Lechner et al., 2023). We apply the same computational lesson to ecological evolution: JAX-based acceleration is not merely an engineering convenience here, but the enabling condition for studying open-ended populations at the scales where the reported behaviors become reliable.

3 METHOD

3.1 ECOLOGICAL GAMES

We formalize the interaction between evolving populations and their environment as an **Ecological Game (EG)**, a multiplayer model designed for objective-free population dynamics. Like Markov decision processes (MDPs) in reinforcement learning and stochastic games (SGs) in multi-agent reinforcement learning (see Sutton et al. (1998) and Albrecht et al. (2024) for reference), an EG has Markovian transition dynamics. Like an SG, it contains a set of agents I , a state space S , an initial state distribution ρ_0 , an action space A , and a transition function $\mathcal{T}$ mapping states s_t and actions for each player $a_{t,i}$ to a distribution over next states s_{t+1} . The key distinction is that an EG removes the reward function and adds a population function $\Gamma(s_t)$, which identifies which agents in I are alive and records parent identities so that births and deaths can be inferred. As with the partially observable counterparts of MDPs and SGs, an EG can also be partially observable; we call this variant a POEG. A POEG restricts agents from observing the full state and instead uses an observation function $O(s_t, i)$ that maps a state and player to that player’s observation. Our environment is a POEG, with observation functions determined by the sensor configurations described below.

3.2 ENVIRONMENT

Our experimental POEG is a large grid world in which agents must collect multiple resources to survive and reproduce. The environment uses a factored state space $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$: s_t^m stores 2D map information such as terrain and resource distributions, while s_t^a stores agent-specific information such as position, health, and internal resources.

Terrain: Each map contains a rock layer that defines grid-cell height. Water initially occupies cells with rock values below a preset sea level, and during an episode it flows downhill according to a discrete flow simulation driven by this height field. We evaluate six global terrain configurations, shown in Figure 2: **Ocean**, **Beach**, **Island**, **Lake**, **Isthmus**, and **Channel**. Experiments span 64×64 , 128×128 , 256×256 , 512×512 , and 1024×1024 grids. When grid size changes, the initial population scales proportionally with area; after initialization, population size is determined entirely by the resulting ecological dynamics.

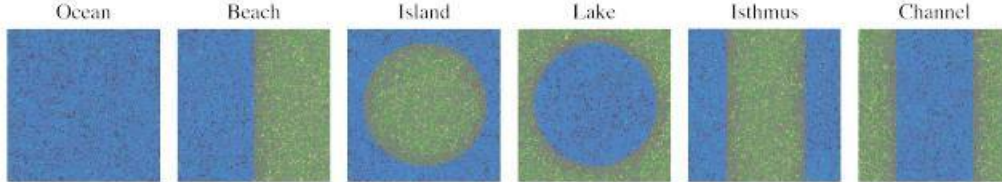


Figure 2: Overhead views of the terrains used in our experiments at 256×256 grid size. Small dark spots are individual agents, and green and yellow regions indicate concentrations of energy and biomass. Each image was captured 5000 steps into an episode, after most biomass in the water had been consumed and before agents had evolved longer-duration survival on land.

Resources and Agent Health: Agents must acquire **water**, **energy**, and **biomass** to remain alive and reproduce. Biomass and energy are initially distributed randomly across the environment, then are consumed, transported, and redeposited through agent behavior.

Every action costs a small amount of energy and water. Spent water returns to the landscape and flows downhill toward nearby bodies of water, while spent energy is destroyed. Biomass enables reproduction: every agent requires a fixed amount of biomass to exist, so reproduction requires enough excess biomass to endow an offspring. Total biomass is conserved throughout a simulation, but unlike water it does not flow or move autonomously. Energy is not conserved; it slowly grows on free-standing biomass. These resource dynamics deliberately create a harder ecological problem than single-food collection because survival depends on managing multiple coupled resources. For ablations on resource rules, see Appendix F.2

Agents also maintain health points (HP), which determine survival. Overspending a critical resource rapidly damages health, while energy and water allow gradual HP recovery. Agents can also attack,

reducing target HP, and they incur a small age-dependent damage term that eventually causes death even in otherwise favorable conditions. When an agent’s HP reaches zero, all carried resources return to the grid cell where it died. Consequently, an agent that kills another can consume the released resources, yielding a direct form of predation. For ablations on HP hyperparameters, see Appendix F.8.

Sensors: Agents observe the world through four sensors. The **internal** sensor reports age, health, and carried resources; the **external** sensor detects resources in the agent’s current grid cell; the **compass**, inspired by animal magnetoreception, reports global direction as a 4-way one-hot vector; and **vision** provides a 7×7 top-down rendered patch centered on the agent, with three color channels and a fourth channel for local elevation difference. These sensors define the information available to evolution, and the experiments below show that seemingly simple sensor differences can decisively change the resulting ecology. For ablations on vision configurations, see Appendix F.6.

Actions: Agents choose from a finite discrete action set grouped into five categories. **Rest** does nothing and has the lowest energy cost. Three **Move** actions rotate clockwise, rotate counterclockwise, or move forward by one grid cell. Because multiple agents cannot occupy the same cell, attempted moves into occupied cells, or conflicts in which multiple agents choose the same destination, are canceled. A single **Attack** action kills agents in the 3×3 square directly in front of the attacker. For ablations on attack strength, see Appendix F.7. Three **Eat** actions correspond to water, energy, and biomass; when the selected resource is present in the agent’s cell, the action transfers a fixed amount of that resource into the agent’s finite-capacity stomach.

The fifth category is **Reproduce**, a single-parent action that creates an offspring. Reproduction succeeds only when the parent has accumulated enough resources to fund the child. The offspring is placed directly behind the parent, inherits a fixed share of the parent’s resources, and receives a mutated copy of the parent’s policy.

Across experiments, we disable selected sensors or the attack action to isolate the ecological consequences of different agent capabilities. We use three sensor combinations: internal+external resources only (**R**), internal+external+compass (**RC**), and internal+external+compass+vision (**RCV**). The suffix **+A** denotes that attacks are enabled; for example, (**RC+A**) agents have resource sensors, a compass, and the ability to attack.

3.3 POLICIES

Agent policies are compact, memoryless MLPs. We intentionally omit memory to keep the computational model simple and the per-agent parameter count low, allowing population scale to be the central experimental variable. For each sensor, raw values are flattened, normalized to $[-1, 1]$, and passed through a separate linear layer with output dimension 64. These sensor embeddings are summed and processed by a 2-layer MLP with hidden dimension 64 and ReLU nonlinearities. Because the action space is discrete, the final layer outputs one logit per action. Actions are sampled from a softmax distribution whose temperature evolves with the network weights. We store weights as half-precision bfloat16 values to support larger populations and faster runtimes. Depending on the active sensors, each agent network contains roughly 10k-25k parameters. For ablations on network size and architecture, see Appendix F.5.

Initial agent weights use Kaiming initialization (He et al., 2015), and bias vectors are initialized to zero. Agents born during an episode receive a copy of the parent’s weights plus Gaussian noise with standard deviation 3×10^{-2} applied to all mutable weight values. This mutation mechanism follows the spirit of prior neuroevolution work (Such et al., 2017) while keeping reproduction deliberately simple. More elaborate operators such as crossover are possible, but the present design isolates environmental scale and ecological interaction without adding unnecessary algorithmic machinery. Empirically, the use of low-precision floating-point values required a larger mutation rate; Appendix F.4 reports the corresponding mutation-rate ablation.

Agents also carry a 3-channel color trait used when rendering them onto the map. This trait mutates like the network weights and determines how an agent appears to other agents with vision. In principle, the color channel gives visual agents a mechanism for distinguishing different types of agents.

3.4 COMPUTATION AND REPRODUCIBILITY

All experiments are implemented in JAX (Bradbury et al., 2018) and run on Nvidia H100 and A40 GPUs in a simulation environment built for large evolving populations. Unless otherwise noted, each experiment runs for 2M environment steps or until extinction, with per-seed wall-clock time ranging from under an hour for 64×64 grids to approximately 10 hours for 1024×1024 grids on one H100 GPU. Appendix E details the simulator architecture, reports scaling measurements, and documents reproducibility considerations.

4 EXPERIMENTS

The experiments demonstrate how sensing modalities, environmental scale, and population size control the behaviors that emerge in objective-free agent populations.

4.1 LONG-DISTANCE RESOURCE GATHERING

The first experiment tests whether agents evolve the ability to travel long distances inland to acquire resources. In environments with clear directional structure, compass-equipped agents (**RC**) consistently develop more effective long-range resource gathering than baseline agents (**R**) without a compass. The behavior also becomes more reliable and more effective as environmental scale increases.

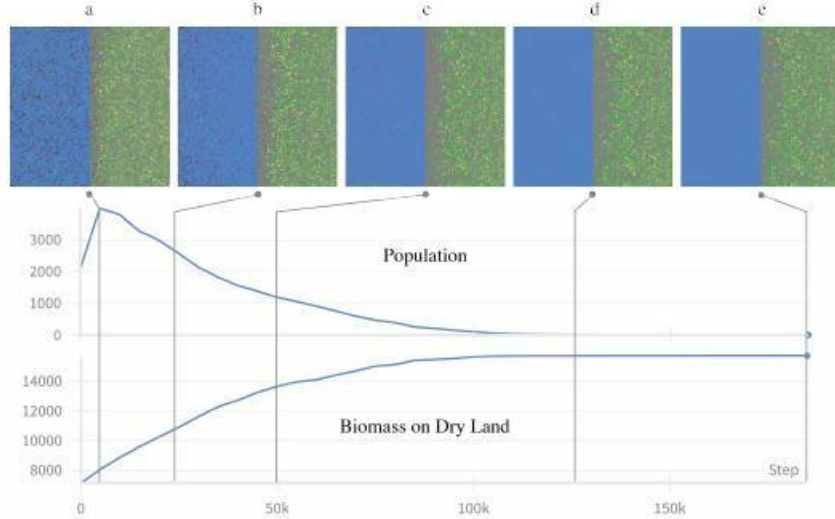


Figure 3: Representative trial with (**R**) agents in a 256×256 **Beach** environment. Population over time appears in the top chart, and free biomass on dry land appears in the bottom chart.

Figure 3 illustrates the failure mode for the (**R**) agent class. After 5000 time steps (a), all initial dry-land agents have died from lack of water, while agents in the water have rapidly expanded by consuming the rich initial resources. At this early stage, the population has not yet evolved policies that efficiently seek resources across terrain boundaries. From the population peak, water-dwelling agents occasionally stumble onto land (b); once there, they typically fail to navigate back to the ocean, die, and deposit biomass on land. By 50,000 time steps (c), this process has nearly doubled the amount of uneaten biomass on land. Because resources have moved from water to inaccessible land, the water population can no longer remain large. By 125,000 steps (d), almost all biomass has shifted onto land, the population is near extinction, and the final agent dies around 185,000 steps (e).

This trajectory shows that navigation failure is not merely an individual-level deficit; it restructures the entire ecology. Agent deaths move biomass into locations the population cannot exploit, reducing carrying capacity and accelerating collapse. At the same time, this resource trap creates a strong opportunity: any lineage that discovers how to enter land, harvest the accumulated biomass, and return safely to water can unlock a new ecological niche.

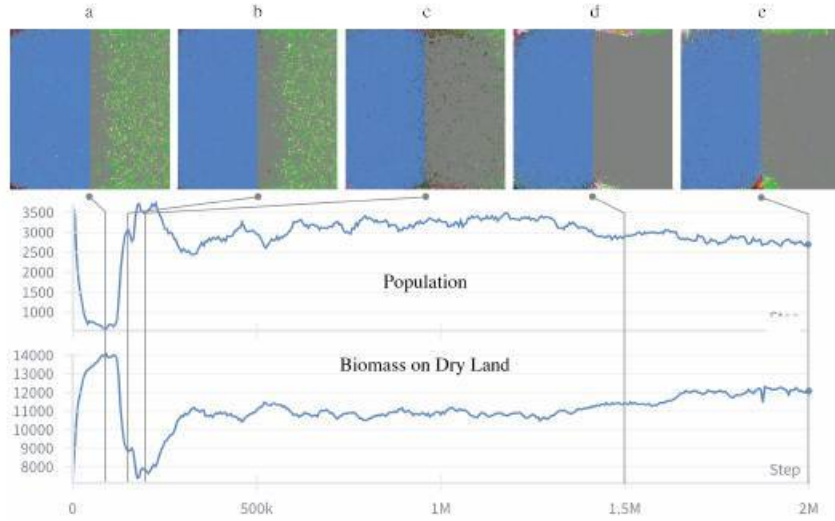


Figure 4: Representative trial with **(RC)** agents in a 256×256 **Beach** environment. Population over time appears in the top chart, and free biomass on dry land appears in the bottom chart.

Figure 4 shows that adding an internal compass **(RC)** changes the outcome. The run begins with the same ecological pressure: the population shrinks and free biomass on land rises (a). However, some agents use the compass to drift toward the relative safety of the corners. Around 150,000 steps (b), agents begin to “mine” the land biomass by navigating back and forth between land and water. By 200,000 steps (c), these miners have cleared the beach. The population later reorganizes into edge clusters (d), but it remains broadly stable through the full 2,000,000 time steps (e).

Table 1: Mining and extinction events in the **Beach** world, comparing compass agents **(RC)** with no-compass agents **(R)** across five world sizes and initial populations. gives the number of seeds with mining, and gives the number of seeds that went extinct before 2M steps. Table 3 in Appendix A reports the seed-level data.

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
	32768		8192		2048		512		128	
(RC)	4/4	0/4	4/4	0/4	4/4	0/4	3/4	1/4	2/4	3/4
(R)	1/4	0/4	0/4	4/4	0/4	4/4	0/4	4/4	0/4	4/4

We quantify this effect with a full set of **Beach** experiments at grid sizes 1024×1024 , 512×512 , 256×256 , 128×128 , and 64×64 , with initial populations of 32768, 8192, 2048, 512, and 128 agents, respectively. For each configuration, we run four random seeds and record whether mining emerges, defined as a drop of at least 10% in free dry-land biomass from its historical maximum. We also record the onset time of mining, defined by the local maximum in land biomass, and the percentage biomass drop. Table 1 summarizes mining and extinction events, and Table 3 in Appendix A reports seed-level details. The results show a clear advantage for **(RC)** agents over **(R)** agents. Mining emerges consistently with a compass, appearing in every seed from the 256×256 scale upward. At smaller scales, populations are less stable and extinction occurs more frequently.

To characterize the timing of these events, we run additional **(R)** and **(RC)** seeds at 512×512 world size (32 seeds) and 128×128 world size (64 seeds), then use a Kaplan-Meier estimator to measure extinction and mining probabilities over simulator time. Extinction is rarer and delayed in larger worlds, while mining is more frequent and appears sooner. A Cox proportional hazards model gives the same qualitative conclusion: larger world size substantially reduces extinction risk and enables

faster mining discovery than smaller worlds, and the compass produces a similar directional effect. Appendix B provides the full analysis.

Table 2: Mining and extinction events in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains, comparing compass agents (RC) with no-compass agents (R) at 512×512 grid size.  gives the number of seeds with mining, and  gives the number of seeds that went extinct before 2M steps. Table 4 in Appendix A reports the seed-level data.

	Island		Lake		Isthmus		Channel	
								
(RC)	4/4	0/4	2/4	2/4	4/4	0/4	3/4	1/4
(R)	3/4	1/4	2/4	4/4	1/4	3/4	0/4	4/4

We next test whether the same mechanism generalizes across terrain structure by running (R) and (RC) agents in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains at 512×512 grid size. Table 2 reports mining and extinction events, while Table 4 in Appendix A gives the seed-level data. As in **Beach**, equipping agents with a compass produces earlier and more consistent mining and lowers extinction. (RC) agents adapt most readily to the island and isthmus terrains, and less reliably to the lake and channel, where some populations still go extinct. The likely reason is structural: in the island and isthmus, walking straight east or west always returns an agent to water, while the channel and lake lack an equally simple return rule. The (R) agents do not reliably adapt in any terrain, although the island is their least difficult case. Figure 12 in Appendix D visualizes mining in the island terrain.

4.2 VISUAL FORAGING AND PREDATION

The second experiment asks whether agents can evolve to use visual information for foraging and predation in the **Ocean** environment. We compare (RCV+A) agents, which have resource sensing, a compass, a 7×7 visual sensor, and attacks, against (RC+A) agents with resource sensing, a compass, and attacks but no vision.

Figure 5 reports population, biomass utilization, movement actions, and eat actions for 512×512 and 128×128 grids. Figure 9 in Appendix C provides the corresponding plots for other grid sizes, and Table 5 in Appendix A summarizes the statistics. Biomass utilization is the percentage of total environmental biomass currently held inside agents. Runs with vision sensors (red) show less movement, more eating, larger populations, and greater biomass utilization. These patterns demonstrate that agents evolve to use vision for more efficient foraging. Scale further strengthens the result: between-run variance decreases substantially in larger environments, and vision-enabled runs exhibit much more stable temporal dynamics across the measured metrics.

Figure 6 reports attacks per agent and homicides per attack for the same runs. Vision-equipped agents attack relatively infrequently but maintain a successful kill rate of approximately 75%, indicating more selective and precise predation. In large grids, blind agents sometimes also achieve high kill rates, but their dynamics are much less stable. The cause of those blind-agent spikes is not fully identifiable from aggregate metrics alone. Visual inspection (Figure 13 in Appendix D) indicates that high blind-agent kill rates can occur when agents cluster densely in one location; such clusters almost never appear in vision runs. This pattern is consistent with vision enabling agents to avoid each other as well as to attack more selectively. In smaller environments, the contrast between visual and blind agents is less pronounced, but visual populations remain far more stable.

4.3 ABLATIONS

Appendix F tests the sensitivity of the emergent behaviors to simulator configuration through ablations on initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7), and agent HP (F.8). We also run the experiments in Appendix F.3 to confirm that the large-environment dynamics are not simply a self-averaging artifact.

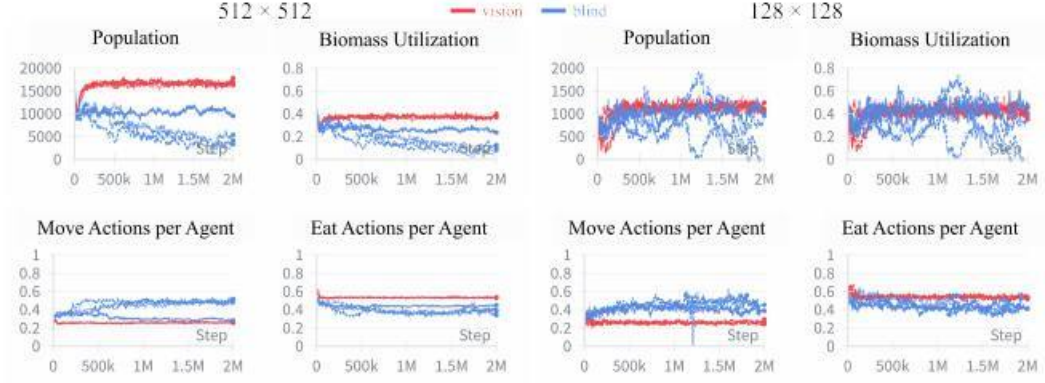


Figure 5: Population, biomass utilization, move actions per agent, and eat actions per agent in 512×512 and 128×128 **Ocean** worlds. Biomass utilization, move actions per agent, and eat actions per agent are averaged across agents at each time step. Each plot shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). Vision produces larger and more stable populations, greater biomass utilization, less movement, and more frequent effective eating. At larger scales, these dynamics become more stable with lower variance across runs, showing that large environments amplify the adaptive value of vision for foraging. See Figure 9 in Appendix C for a comparison of grid sizes 1024×1024 and 256×256 .

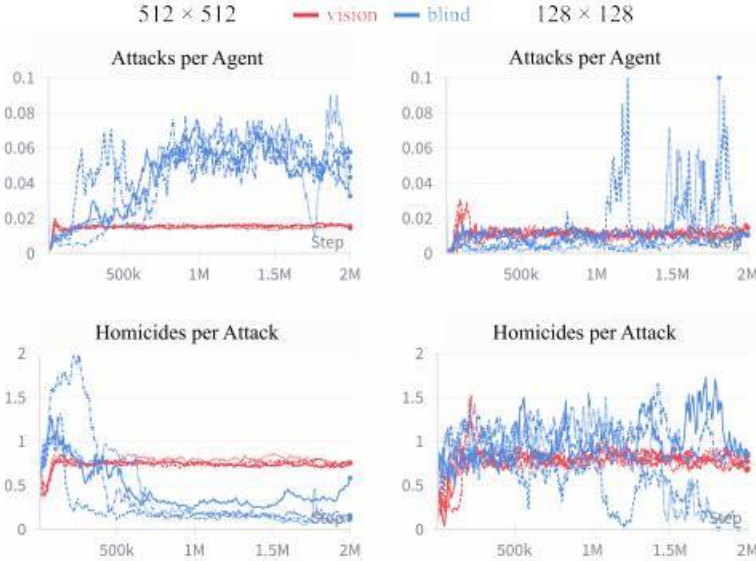


Figure 6: Attacks per agent and homicides per attack, averaged across agents, in 512×512 and 128×128 **Ocean** worlds. Each plot shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). Vision leads agents to attack less often but with substantially higher precision, supporting specialized predation behavior. This separation is strongest at larger scales. See Figure 10 in Appendix C for a comparison of grid sizes 1024×1024 and 256×256 .

5 CONCLUSION

This study establishes open-ended ecological simulation as a powerful paradigm for investigating the emergence of intelligent behavior. Across worlds that vary in scale, terrain, and sensory affordances, we show that environmental scale and sensory richness materially shape the evolution of sophisticated sensing and decision-making policies. The results demonstrate that ecological and evolutionary pressures, when paired with rich environments, can produce complex behavior without explicit objectives. They also show that large populations of neural agents can be evolved within a

practical hardware budget. Together, these findings make a clear case for ecology as both a mechanism for developing artificial agents and an experimental lens for understanding how intelligence can emerge from competition, reproduction, and environmental feedback.

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A ADDITIONAL DATA

Table 3: Seed-level mining events in the **Beach** world, comparing compass agents (**RC**) with no-compass agents (**R**) across five scales. 🛠️ marks runs with a mining event. 📉 reports the percent decrease in free dry-land biomass during the mining event, and ⌚ reports the approximate timestep at which free dry biomass reached the local maximum that marks mining onset. For runs with multiple mining events, we report the first event. 💀 reports the extinction timestep, or n/a for runs that lasted 2M steps.

	1024x1024				512x512				256x256				128x128				64x64			
	32768				8192				2048				512				128			
Compass On (RC)																				
Seed																				
0	✓	35k	11.1	n/a	✓	105k	33.	n/a	✓	145k	49.3	n/a	✓	105k	72.5	n/a	x			55
1	✓	90k	16.2	n/a	✓	50k	17.2	n/a	✓	50k	46.6	n/a	x		140	✓	70k	85.3	675	
2	✓	40k	15.	n/a	✓	30k	14.3	n/a	✓	55k	34.7	n/a	✓	50k	50.7	n/a	✓	10k	78.	n/a
3	✓	60k	15.	n/a	✓	45k	15.6	n/a	✓	90k	47.4	n/a	✓	20k	52.	n/a	x			135
Avg	4/4	34k	14.3	0/4	4/4	58k	20.	0/4	4/4	85k	44.5	0/4	3/4	58k	58.4	1/4	2/4	40k	81.7	3/4
Compass Off (R)																				
Seed																				
0	x			n/a	x			945k	x			185k	x			65k	x			45k
1	✓	275k	12.4	n/a	x			1085k	x			110k	x			55k	x			40
2	x			n/a	x			1070k	x			165k	x			55k	x			25
3	x			n/a	x			280k	x			125k	x			90k	x			40
Avg	1/4	275k	12.4	0/4	0/4			n/a	0/4			4/4	0/4			4/4	0/4			4/4

Table 4: Seed-level mining events in **Island**, **Lake**, **Isthmus**, and **Channel** terrains, comparing compass agents (**RC**) with no-compass agents (**R**). Each run uses a 512×512 grid with initial population 8192.

	Island				Lake				Isthmus				Channel			
Compass On (RC)																
Seed																
0	✓	40k	94.3	n/a	x			45k	✓	45k	33.	n/a	✓	50k	27.4	n/a
1	✓	30k	94.3	n/a	✓	30k	22.5	n/a	✓	55k	37.3	n/a	✓	80k	30.8	n/a
2	✓	30k	94.8	n/a	x			45k	✓	40k	28.1	n/a	✓	50k	41.	n/a
3	✓	70k	95.1	n/a	✓	35k	25.3	n/a	✓	30k	41.5	n/a	x			80k
Avg	4/4	43k	94.6	0/4	2/4	33k	23.9	2/4	4/4	43k	35.	0/4	3/4	60k	33.1	1/4
Compass Off (R)																
Seed																
0	x			185k	x			125k	x			145k	x			100k
1	✓	75k	30.9	n/a	x			130k	x			260k	x			105k
2	✓	85k	47.	n/a	✓	75k	23.4	1400k	✓	95k	18.8	n/a	x			105k
3	✓	115k	39.5	n/a	✓	70k	24.3	1250k	x			140k	x			100k
Avg	3/4	92k	39.1	1/4	2/4	73k	23.9	4/4	1/4	95k	18.8	3/4	0/4			4/4

Table 5: Population and foraging metrics for Vision and Blind agents in the **Ocean** world across scales. denotes population size. denotes biomass utilization, the percentage of total system biomass currently inside agent stomachs. denotes move actions per agent, averaged across agents, and denotes the proportion of actions that are eat actions. Reported , , , and values are seed-level averages over all steps.

	1024x1024 32768				512x512 8192				256x256 2048				128x128 512			
Vision (RCV+A)																
Seed																
0	58354	.396	.402	.426	16413	.369	.257	.532	4195	.382	.255	.536	1126	.426	.261	.541
1	65107	.365	.257	.529	16200	.364	.256	.533	4142	.377	.258	.536	1103	.415	.262	.537
2	64735	.363	.257	.530	16560	.372	.257	.533	4269	.388	.260	.536	1041	.392	.254	.542
3	64595	.362	.256	.532	16206	.365	.255	.531	4249	.388	.259	.533	1100	.414	.257	.538
Avg	63198	.371	.293	.504	16345	.368	.256	.532	4213	.384	.258	.535	1092	.412	.259	.539
Blind (RC+A)																
Seed																
0	22734	.143	.446	.372	6549	.173	.442	.400	2036	.209	.433	.461	545	.219	.465	.427
1	23670	.151	.426	.388	6939	.180	.438	.396	3107	.321	.390	.416	1082	.444	.416	.460
2	24745	.159	.358	.426	5448	.136	.474	.374	1587	.163	.440	.433	979	.421	.404	.436
3	20004	.130	.469	.359	10354	.265	.311	.439	3164	.324	.321	.450	788	.317	.431	.477
Avg	22788	.146	.425	.386	7323	.189	.416	.402	2473	.254	.396	.440	848	.351	.429	.450

Table 6: Predation metrics for Vision and Blind agents in the **Ocean** world across scales.  denotes attacks per agent, the fraction of actions that are attacks averaged across agents.  denotes homicides per attack, the number of agents killed per attack action. Reported  and  values are seed-level averages over all steps. The 0.562  value for seed 3 in the 128x128 (**RC+A**) run is an extinction-driven outlier: the seed went extinct before 2M steps, causing attacks per agent to spike as population size dwindled.

	1024x1024		512x512		256x256		128x128	
	32768		8192		2048		512	
Vision (RCV+A)								
Seed								
0	.011	.776	.015	.731	.013	.784	.010	.788
1	.016	.733	.015	.741	.014	.780	.011	.793
2	.016	.707	.015	.737	.014	.750	.012	.771
3	.016	.712	.015	.780	.013	.759	.011	.765
Avg	.015	.732	.015	.747	.014	.758	.011	.779
Blind (RC+A)								
Seed								
0	.070	.218	.042	.437	.005	1.074	.016	.558
1	.054	.300	.044	.383	.016	.871	.006	1.032
2	.076	.236	.048	.238	.031	.383	.010	.979
3	.059	.203	.044	.456	.033	.446	.562	.841
Avg	.065	.239	.045	.378	.021	.693	.148	.853

B DEMOGRAPHY

We directly compare extinction and mining emergence as functions of world size and compass access. Extinction occurs when the population reaches zero individuals, and mining uses the same definition as in the main experiments: a 10% drop in free biomass on land.

We group runs by world size and compass access. We simulated 64 runs each for 128×128 without a compass and 128×128 with a compass, and 32 runs each for 512×512 without a compass and 512×512 with a compass. For each group, we fit a non-parametric Kaplan–Meier estimator to the time-to-extinction distribution, treating runs that did not go extinct within 2M steps as right-censored. Figures 7 and 8 show the central demographic result: larger worlds and compass access both extend survival and accelerate mining emergence.

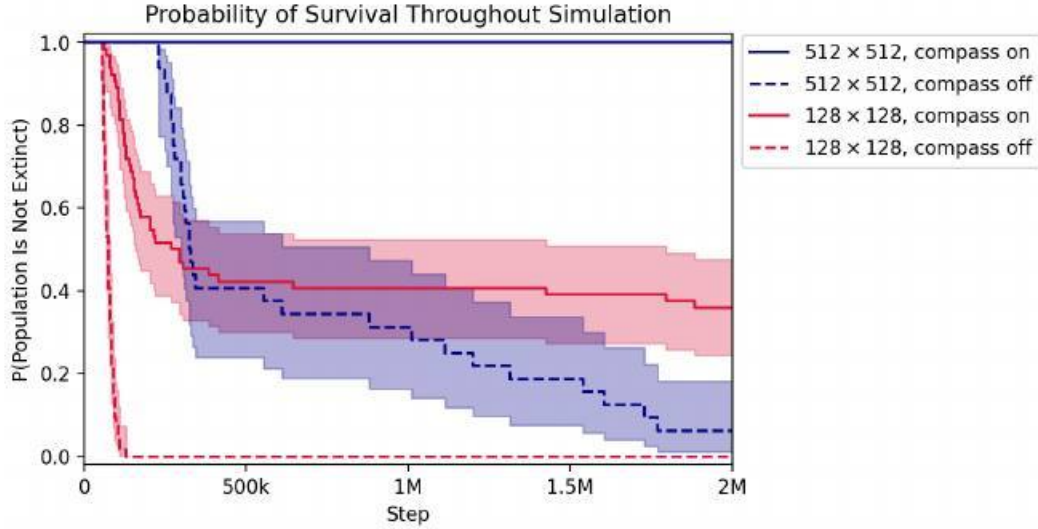


Figure 7: Survival probability as a function of world size, compass access, and timestep. We fit one non-parametric Kaplan–Meier estimator for each run group (e.g. 512×512 , compass on). Shaded regions give 95% confidence intervals. Each 512×512 group uses 32 runs, and each 128×128 group uses 64 runs.

We quantify emergence and extinction timing with a Cox proportional hazards model, a semi-parametric survival model relating covariates to event hazards (Table 7). Runs without an emergence or extinction event by 2M time steps are treated as censored. For extinction, larger world size and compass access both sharply reduce hazard: the extinction rate in 128×128 worlds is about 25 times greater than in corresponding 512×512 worlds ($HR=0.04$), and enabling the compass has a similarly large effect ($HR=0.03$). For mining, larger worlds produce much faster emergence ($HR=6.07$), and compass access accelerates emergence dramatically ($HR=34.59$). The models discriminate well between groups, with concordance 0.80 for extinction and 0.85 for mining. These results support a clear mechanism: larger worlds sustain larger populations, lower the probability of reaching the absorbing zero-population boundary, and simultaneously explore a broader phenotype space through more mutation events than smaller worlds.

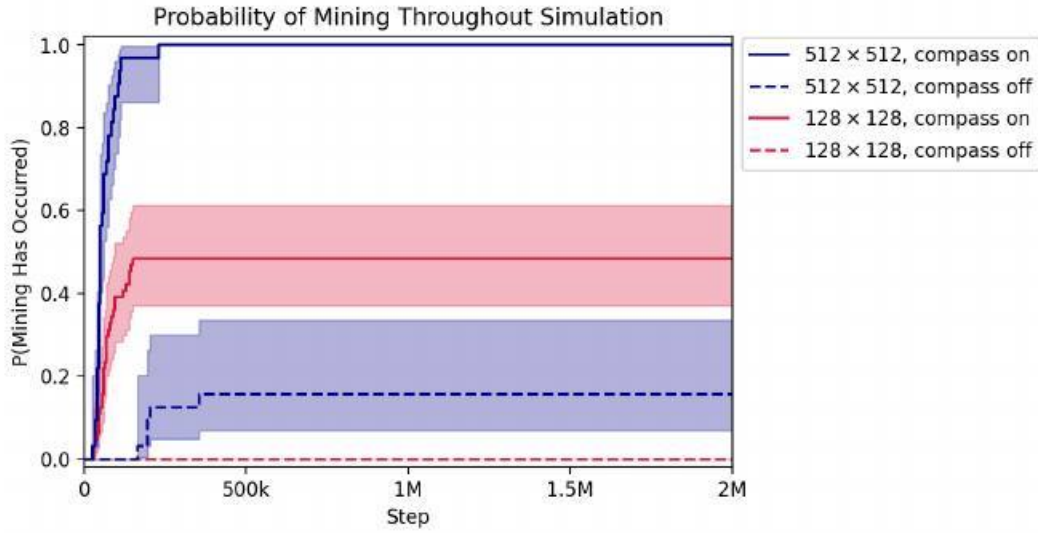


Figure 8: Probability that mining has emerged as a function of world size, compass access, and timestep. We fit one non-parametric Kaplan-Meier estimator for each run group (e.g. 512×512 , compass on). Shaded regions give 95% confidence intervals. Each 512×512 group uses 32 runs, and each 128×128 group uses 64 runs.

Table 7: Cox proportional hazards models for extinction and emergence events with a 128×128 , compass off baseline. Hazard ratios (HR) less than 1 indicate reduced hazard (longer time to event); for example, extinction $HR = 0.04$ for 512×512 means extinction occurs at 0.04 times the rate of 128×128 . There are 192 total runs: 32 seeds with compass on and 32 seeds with compass off in the 512×512 world, and 64 seeds with compass on and 64 seeds with compass off in the 128×128 world.

Covariate	Extinction			Mining		
	HR	95% CI	p-value	HR	95% CI	p-value
Size 512×512	0.04	0.02–0.07	< 0.005	6.07	3.61–10.20	< 0.005
Compass On	0.03	0.01–0.05	< 0.005	34.59	13.51–88.57	< 0.005
Concordance	0.80			0.85		
Events / Total	135 / 192			68 / 192		

C ADDITIONAL FORAGING AND PREDATION RESULTS

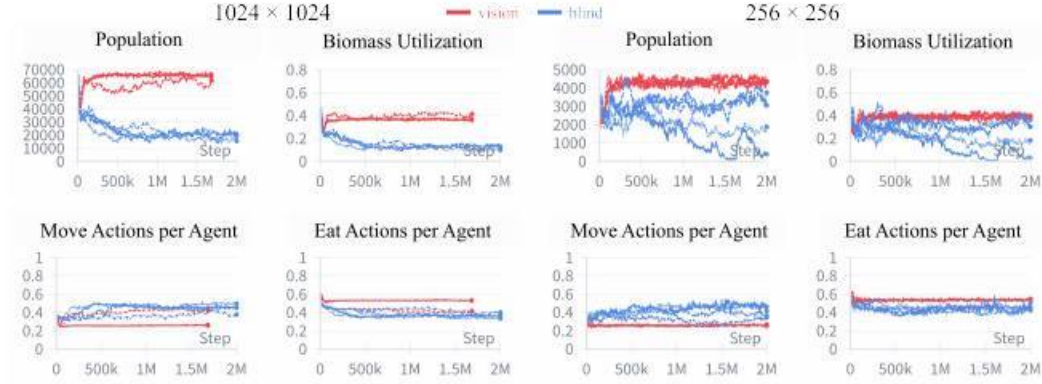


Figure 9: Foraging behavior in the **Ocean** world across scales. Vision agents (red) achieve more stable populations and higher biomass utilization by eating more effectively and moving less. Larger scales further stabilize these dynamics and reduce variance across runs. Due to GPU-hour limits, the vision seeds at the 1024×1024 scale ran for 1.7M rather than 2M steps.

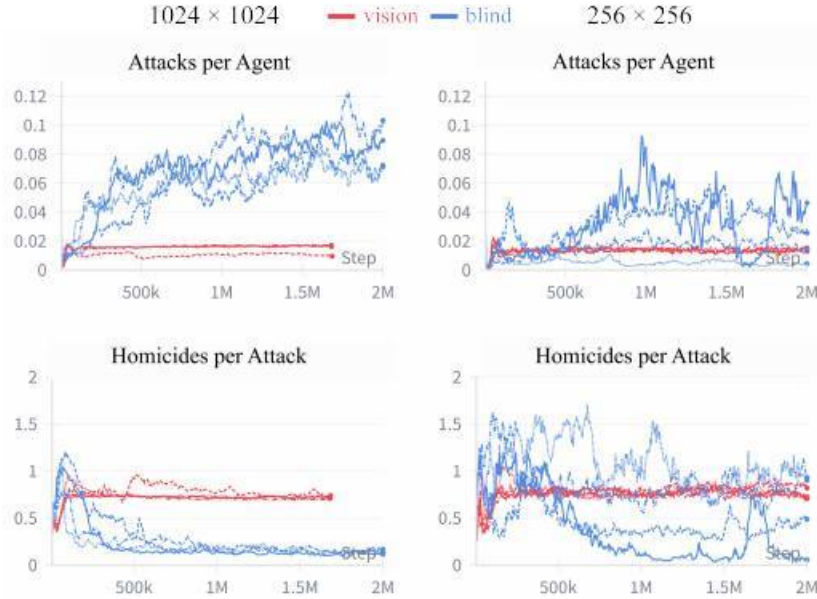


Figure 10: Predation behavior in the **Ocean** world across scales. Vision agents (red) attack less frequently but with much higher precision, demonstrating specialized predation alongside stronger foraging (Figure 9). The separation is strongest at larger scales, while blind agents (blue) show greater behavioral variance in smaller worlds.

D ADDITIONAL EXPERIMENT VISUALIZATIONS

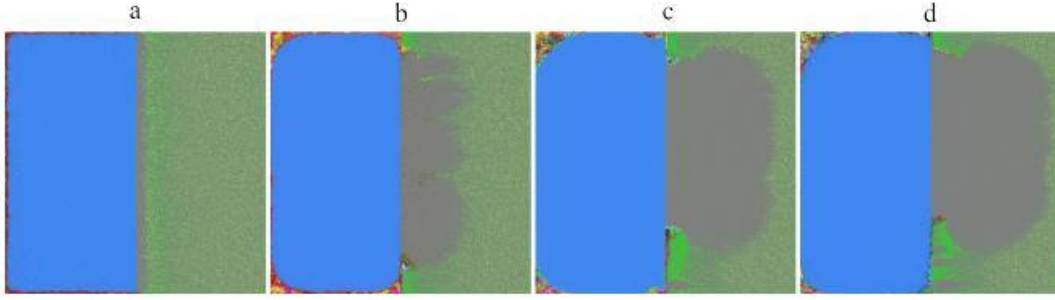


Figure 11: Large-scale environments produce varied, complex ecologies with distinctive spatial patterns. In this 1024×1024 **Beach** world, **(RC)** agents initially die at the shore (a), transferring biomass to dry land. They then evolve mining behavior and clear biomass from the beach (b). When biomass re-accumulates at the beach ends (c), subpopulations re-establish mining behavior and travel inland along these smaller biomass beaches to extract resources (d).

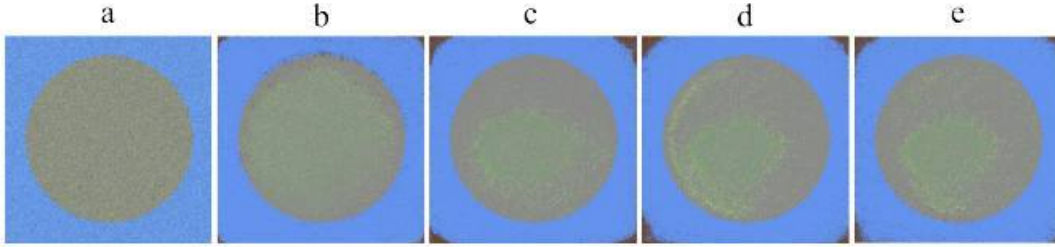


Figure 12: Mining behavior emerges across varied terrains. In this 512×512 **Island** world with **(RC)** agents, different shore regions are mined over the run (b) (c). When mining becomes difficult and agents die before returning to water, biomass is transferred back onto parts of the island (d). Agents then adapt to re-mine this accumulated biomass wall, clearing it from the shore (e).

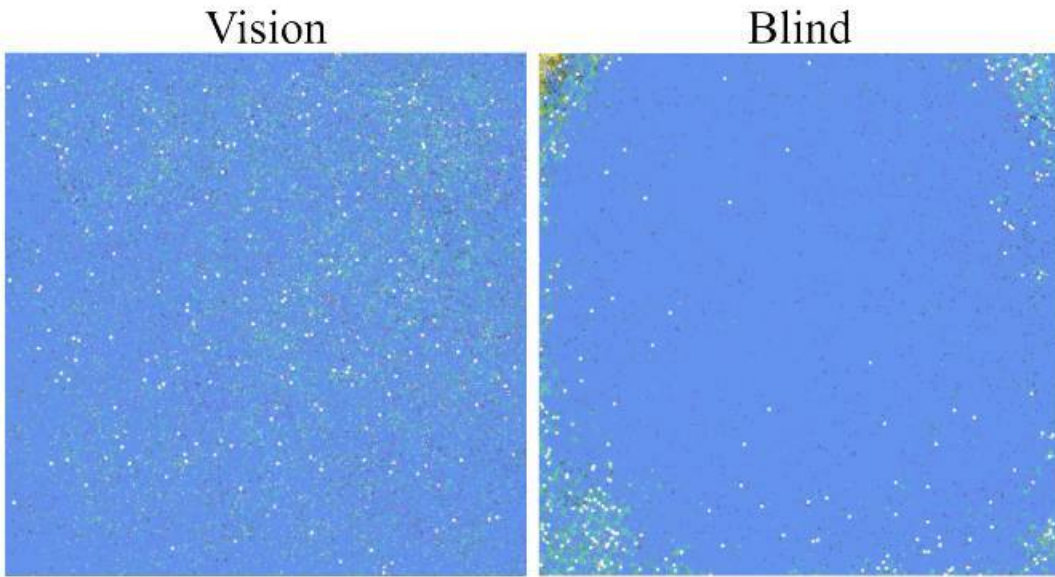


Figure 13: Representative attacking-enabled runs in a 512×512 **Ocean** world, with vision **(RCV+A)** agents on the left and blind **(RC+A)** agents on the right. White squares mark attack locations. Vision agents adapt to locate free resources throughout the ocean and prey more effectively on other agents (see Figures 5 and 6). Blind agents explore the ocean much less and instead evolve corner-clustering policies.

E DETAILS ON COMPUTATION AND REPRODUCIBILITY

We provide implementation details that support the paper’s scale claims, including simulator design, runtime, and reproducibility.

Simulator design. All experiments use a JAX-based, GPU-accelerated simulator built specifically for large-scale multi-agent ecosystems with open-ended populations. The simulator computes every component of an environment step on the GPU—terrain dynamics, resource flows, agent updates, and measurement—so policy evaluation and environment simulation share a single JAX compilation pipeline. It exposes a minimal population-level interface: given map state, agent state, and heritable traits, a population algorithm supplies actions and updated traits while the simulator handles births, deaths, and environmental transitions. This separation makes the same environment implementation compatible with evolutionary algorithms, reinforcement learning methods, and other population update rules. Pure JAX functions, vectorization, and just-in-time compilation fuse operations into a small number of GPU kernels, delivering the throughput required for the scaling experiments reported in Table 8.

The simulator also provides built-in tooling that is essential for interpreting emergent behavior. Its measurement API records aggregate statistics such as biomass utilization, movement and eating rates, attack and homicide rates, and environment-level parameters at every step; these measurements feed directly into the figures and tables. An interactive 3D viewer renders terrain, water, resources, and agents from generated reports and supports inspection of individual agents (Figure 1). These tools make qualitative phenomena such as long-range mining and predation patterns observable and verifiable when scalar metrics alone are insufficient.

From a high-performance computing perspective, the simulator fills a clear gap. General-purpose agent-based frameworks such as MASON (Luke et al., 2005), FLAME-GPU (Richmond et al., 2024), or Griddly (Bamford et al., 2020) are configurable but require substantial environment engineering and do not directly target open-ended evolving populations at the scales studied here. GPU-native multi-agent RL environments, in contrast, typically assume fixed agent counts and task-specific rewards. Our simulator combines GPU-native execution, explicit births and deaths, mutable heritable traits, and configurable resource and terrain dynamics, enabling open-ended populations of over 10^5 agents while remaining interoperable with multiple population algorithms. This combination makes it a state-of-the-art platform for scaling ecological experiments and a strong foundation for future studies of emergent behavior in large populations. Our simulator will be released upon acceptance.

Runtime data. Table 8 reports runtime across grid sizes and maximum agent populations. Each run uses a single H100 GPU for at most 2M steps. The table reports (RC) and (RCV + A) agents with 2 MLP layers and 64 channels per layer ($\sim 22k$ parameters total); runtime also depends on network size, active sensors, and logging level.

At the 1024×1024 scale, the largest runs complete in ~ 10 hours for **Beach** and ~ 14 hours for **Ocean** on one H100, with memory usage of roughly 60GB. **Ocean** runs are slower because they include both attack actions and vision sensing.

Table 8: Runtime in the **Beach** and **Ocean** worlds across grid sizes and maximum agent populations. Each (RC) agent has a 22,465-parameter neural network, and each (RCV + A) agent has a 22,529-parameter neural network. We report simulation steps per second (hz) and extrapolated wall-clock runtime in hours for 2M steps. Effective runtime is lower for seeds that go extinct before 2M steps.

Environment	1024×1024		512×512		256×256		128×128		64×64	
	hz	runtime	hz	runtime	hz	runtime	hz	runtime	hz	runtime
Beach (RC)	55	10.1 hrs	190	2.9 hrs	315	1.8 hrs	535	1.0 hrs	750	0.7 hrs
Ocean (RCV + A)	40	13.9 hrs	142	3.9 hrs	235	2.4 hrs	415	1.3 hrs	750	0.7 hrs

Reproducibility. Because the environments are large, randomized, and sensitive to long-horizon dynamics, individual runs can diverge in precise outcomes. We therefore report four random seeds

for each main experiment. Repeating an experiment with the same random seed does not reliably reproduce an identical trajectory because some low-level GPU operations are nondeterministic: parallel floating-point reductions can execute in different orders, and fused operations can change rounding behavior. Tiny finite-precision differences can accumulate over many steps into visible trajectory drift. The use of 16-bit weights can amplify rounding effects but is not the primary source of nondeterminism. Despite this run-level variability, the aggregate results across seeds reveal stable high-level patterns.

F ABLATIONS

We present ablations that test how specific hyperparameters and environment or agent configurations affect the robustness of the main emergent behaviors.

F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

We first isolate the effect of initial population size in a 256×256 **Beach** world, measuring both mining consistency and population stability for (**RC**) agents. Agents without a compass (**R**) provide a sharp contrast: they had 0% mining and 100% extinction across all initial population sizes except 8192, the largest setting, where they reached 25% mining and 75% extinction. Mining and extinction frequencies need not sum to 100% because a population can mine and later go extinct, or survive for the full 2M steps without mining.

Figure 14 shows that initial population size does not materially change mining consistency or extinction rates in this setting: populations initialized with only 128 agents evolve mining as often as populations initialized with 8192 agents, and extinction rates remain largely stable across the sweep.

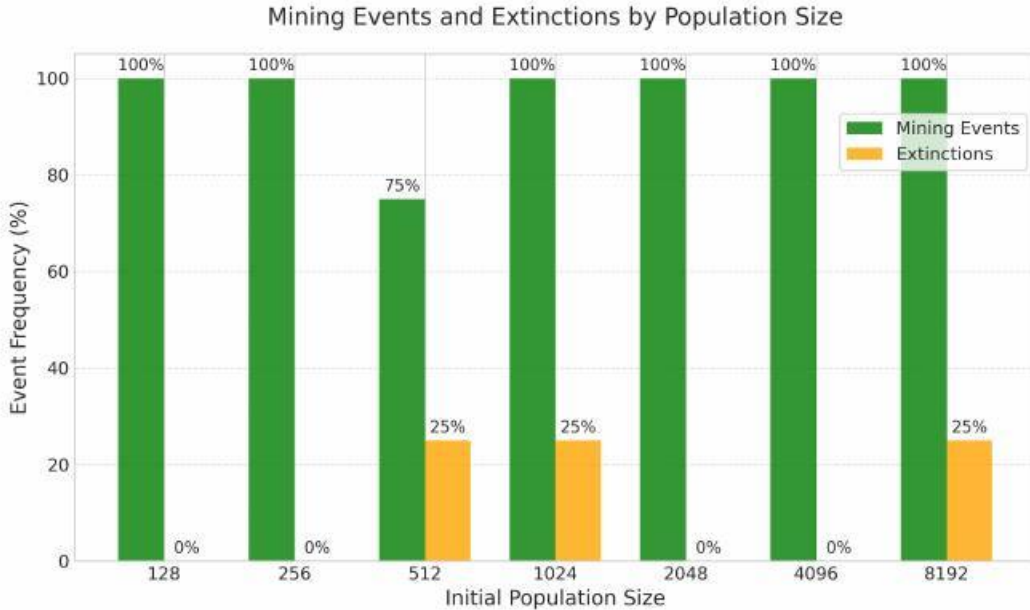


Figure 14: Frequency of mining and extinction events in a 256×256 **Beach** world across initial sizes of (**RC**) populations. Frequencies use four random seeds per initial population size; for example, 25% Extinction Event Frequency means the population reached zero before 2M simulation steps in 1/4 seeds.

Figure 15 explains why the initial population has limited downstream effect. Regardless of initialization, each run rapidly approaches a similar population size within the first 50,000 simulation steps. The environment therefore imposes an implicit carrying capacity determined by fixed system resources: populations below capacity grow toward it, and populations above capacity are quickly

constrained by it. After this early correction, effective population sizes are similar enough that mining and extinction frequencies remain stable. Larger initial populations still add some resources because agents carry biomass that is deposited at death, and therefore they reach somewhat higher maximum populations on average (Table 9); however, this increase is not large enough to change mining or extinction frequency in this setting. Later population trajectories can still diverge substantially because agents evolve different capabilities at different times, reshaping resource distributions in distinct ways.

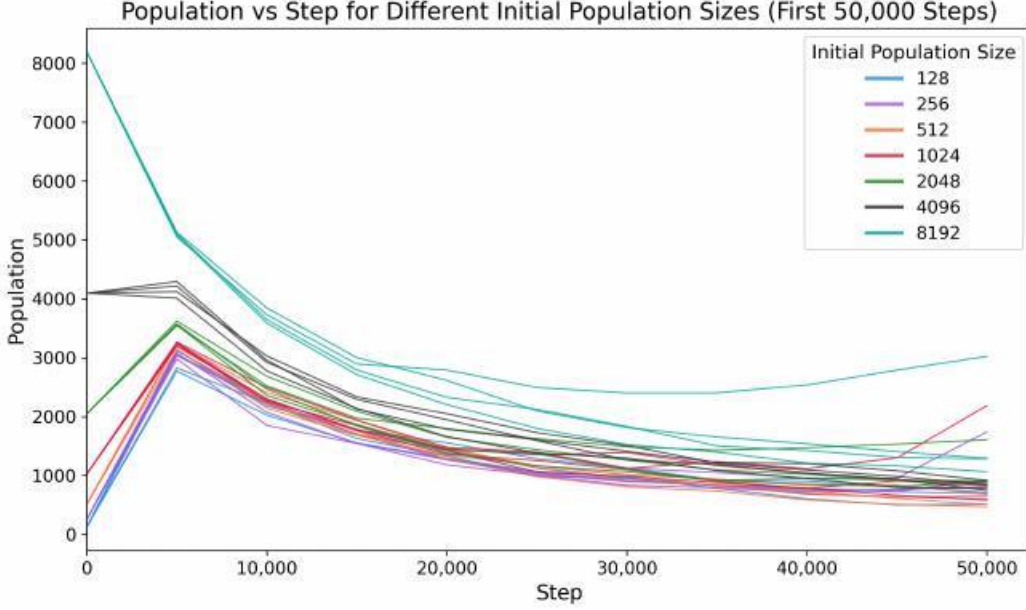


Figure 15: Population size during the first 50,000 time steps in the 256×256 **Beach** with (RC) agents, using four seeds per initial population size. The environment enforces an implicit carrying capacity through available resources, and all initial population sizes approach this value early in the run. Larger initial populations contribute additional biomass through agent deaths, modestly raising the carrying capacity.

Table 9: Maximum post-initialization population size for each starting population, reported as averages and standard deviations across four random seeds. Larger initial populations reach slightly higher maxima and show greater variance, but the maximum population is largely robust to initialization: a starting population of 128, only 1.6% of 8192, still reaches 78.9% of the maximum reached by the 8192 setting (3178 vs. 4026).

Initial Population Size	Max Population Size After Step 0
128	3178 ± 217
256	3056 ± 61
512	3451 ± 437
1024	3613 ± 381
2048	3786 ± 431
4096	3689 ± 437
8192	4026 ± 642

F.2 VARYING RESOURCE RULES

We next test how two resource hyperparameters—biomass photosynthesis rate and initial biomass site density—control mining and extinction frequency. The setting matches Appendix F.1 with initial population fixed at 2048 and compass-equipped (RC) agents.

The biomass photosynthesis rate sets how quickly landscape biomass creates energy, providing renewable resources over time. The initial biomass site density sets the fraction of grid cells containing biomass at simulation start. All main experiments (Section 4) use a biomass photosynthesis rate of 0.01 and an initial biomass site density of 1/4.

Unlike initial population size, resource rules strongly affect mining and extinction consistency (Figures 16 and 17).

Higher biomass photosynthesis rates and higher initial biomass site densities directly raise effective carrying capacity by increasing available resources. Larger sustained populations then create more opportunities for mutations that support mining while reducing extinction risk. The data support this mechanism: omitting extinct seeds, runs with initial biomass site density 1/16 reached an average post-step-0 maximum population of 1608, compared with 2466 for density 1/8, 3786 for density 1/4, and 5353 for density 1/2.

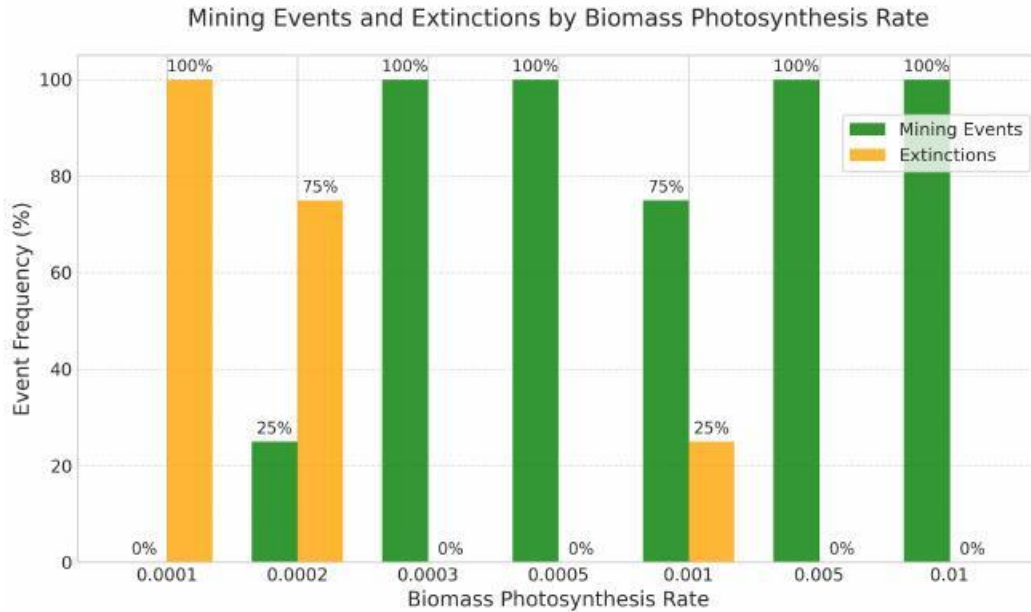


Figure 16: Frequency of mining and extinction events in a 256×256 **Beach** world across biomass photosynthesis rates. Frequencies use four random seeds per setting. Mining emerges consistently once biomass photosynthesis reaches 0.0003 or greater.

F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

One possible alternative explanation is that the increased stability of mining in large environments (Table 1) reflects self-averaging rather than genuinely different dynamics. By self-averaging, we mean the statistical-physics intuition that, as system size grows, fluctuations in macroscopic quantities shrink roughly as $1/\sqrt{N}$, where N is the system size, so a single large system behaves like the average of many smaller systems with the same total size. Under that hypothesis, a 512×512 **Beach** world would be equivalent to many loosely coupled 128×128 worlds run in parallel, and our scaling results could be reproduced by averaging over enough small-world seeds.

We test this alternative directly by holding total “sample size” roughly fixed while changing how it is partitioned across environments. In the **Beach** terrain with compass-equipped agents (**RC**), we compare three conditions:

- 4 runs of a 512×512 world with 8192 initial agents each,
- 16 runs of a 256×256 world with 2048 initial agents each,
- 64 runs of a 128×128 world with 512 initial agents each.

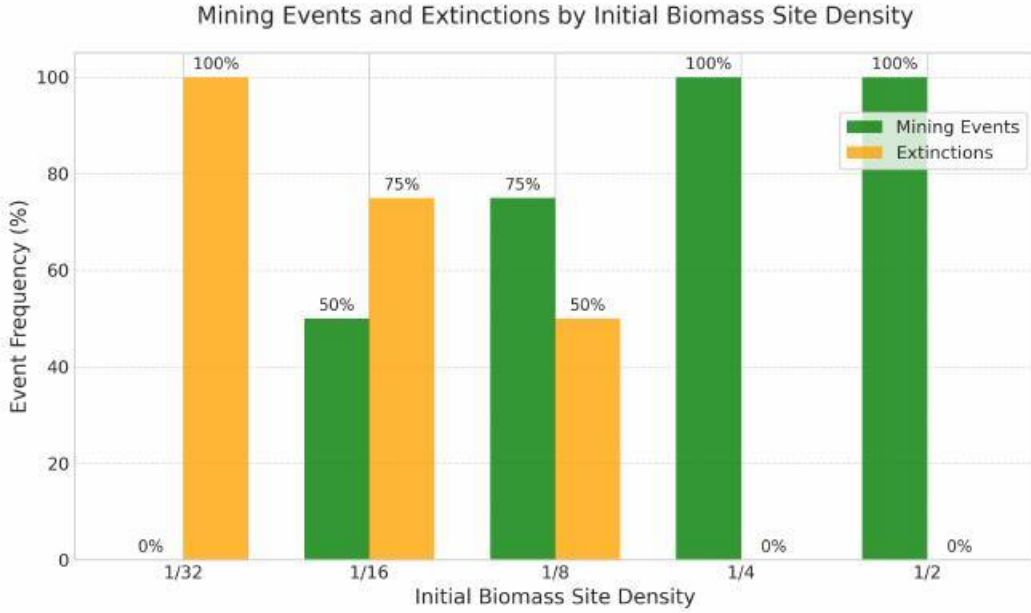


Figure 17: Frequency of mining and extinction events in a 256×256 **Beach** world across initial biomass site densities. Frequencies use four random seeds per setting. Higher initial biomass site density produces more consistent mining discovery and lower extinction risk.

In each case, the total number of grid cells across runs, the total starting resources and the total initial number of initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed runs for 2M steps with the same hyperparameters as in Section 4.1. As a control, we repeat the protocol for agents without a compass (**R**). Table 10 summarizes long-distance mining and extinction frequencies in each configuration.

If large environments were only self-averages of smaller ones, mining and extinction rates would be roughly invariant to how grid cells and agents are partitioned across runs: 4 large worlds and 64 small worlds would have similar mining probabilities. The data decisively reject that pattern. For (**RC**) agents, mining appears in every 512×512 run with no extinctions, remains common but not guaranteed at 256×256 , and becomes unreliable at 128×128 , where many populations go extinct before 2M steps. In contrast, (**R**) agents never mine and go extinct at every scale, confirming that the compass is a key enabler of long-range behavior.

Table 10: Mining and extinction events when world size is traded against the number of independent runs. We compare **Beach** worlds at 512×512 , 256×256 , and 128×128 with proportionally scaled initial populations (8192, 2048, and 512 agents). Each condition allocates the same total number of grid cells and initial agents across all runs by using 4 seeds at 512×512 , 16 seeds at 256×256 , and 64 seeds at 128×128 . We report the fraction of runs with mining (🔍) and extinction before 2M steps (💀) for compass agents (**RC**) and no-compass agents (**R**).

	512 × 512 8192		256 × 256 2048		128 × 128 512	
	🔍	💀	🔍	💀	🔍	💀
(RC)	4/4	0/4	15/16	1/16	31/64	41/64
(R)	0/4	4/4	0/16	16/16	0/64	64/64

These results rule out self-averaging as the sole explanation. Larger environments enable and stabilize mining in ways that cannot be recovered simply by averaging many small simulations.

Even with the same total number of cells, agents, and simulation steps, concentrating resources into a few large worlds creates a regime where mining is consistent and extinction is rare. Dispersing the same resources across many small worlds instead yields populations that frequently fail to discover mining and often collapse. Increasing environmental scale therefore changes the underlying ecological dynamics, plausibly by supporting larger, more spatially distributed populations and more persistent diversity in evolving policies, rather than merely smoothing stochastic noise.

F.4 MUTATION RATE SWEEP

We test how sensitive mining emergence is to mutation rate, the hyperparameter controlling the scale of Gaussian noise added to an offspring policy at birth. Mutation rate directly determines how far child policies deviate from parent policies. We again use a 256×256 **Beach** world with an initial population of 2048 (**RC**) agents and run for at most 2M steps.

Figure 18 shows that for mutation rates from 6×10^{-3} to 7.875×10^{-2} , mining emerges in at least 3 of 4 seeds. The main mutation rate, 3×10^{-2} , therefore lies inside a broad robust interval. Mutation rates that are too small, such as 3×10^{-3} , prevent useful behavioral innovation and impair survival, leading to extinction. Mutation rates that are too large, such as 3×10^{-1} , destabilize populations and consistently produce extinction without mining adaptation.

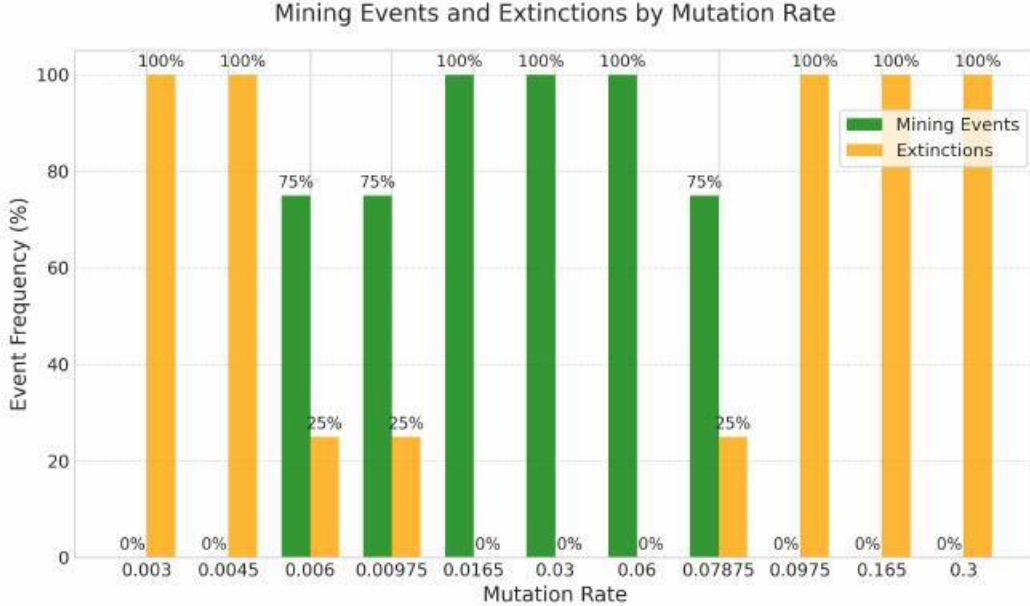


Figure 18: Frequency of mining and extinction events in a 256×256 **Beach** world for different mutation rates. Frequencies are computed across four random seeds for each setting.

F.5 NETWORK SIZE

We study the effect of the size and architecture of the agent policy network on mining and extinction frequency, as well as the impact of environmental scale on this effect. We investigate 256×256 and 512×512 **Beach** worlds with initial population sizes of 2048 and 8192 (**RC**) agents, respectively (scaling by four with the number of grid cells). We ran baselines with (**R**) agents for each experiment, again finding rare to no mining events and 100% extinction rates.

The main experiments in Section 4 use 2-layer, 64-channel policies with 22465 parameters per agent (Table 11). Figure 19 shows that, in the 256×256 **Beach**, halving channels to 32 (9185 parameters) or doubling them to 128 (61313 parameters) while keeping depth fixed has little effect on mining

or extinction frequency. Network depth matters more: all populations with 1 layer and 32 channels went extinct, and 50% of populations with 3 layers and 128 channels went extinct.

Crucially, scaling to 512×512 changes the network-size picture (Figure 20). In the larger world, populations with 3-layer, 128-channel networks consistently mine and survive through 2M steps. Populations with small 1-layer, 32-channel networks also mine consistently. This result reinforces the central scale claim: larger environments make behavior more robust to policy-network variation. Small networks may struggle because reduced capacity limits available solutions, while large networks may require longer search because the parameter space is larger. In smaller worlds, these difficulties often lead to failure; in larger worlds, the population and spatial scale provide enough exploration and ecological buffering for successful lineages to appear and persist. Larger environments support more policy-space exploration, and locally successful mutants propagate more slowly across space, preserving diversity for longer. For more discussion of these effects see (Mitchell et al. 2006).

Table 11: Total parameters in (RC) and (R) agent policies with different network architectures.

Architecture (layers, channels)	Parameter Count (RC)	(R)
1, 32	8129	7969
2, 32	9185	9025
2, 64	22465	22145
2, 128	61313	60673
3, 128	77825	77185

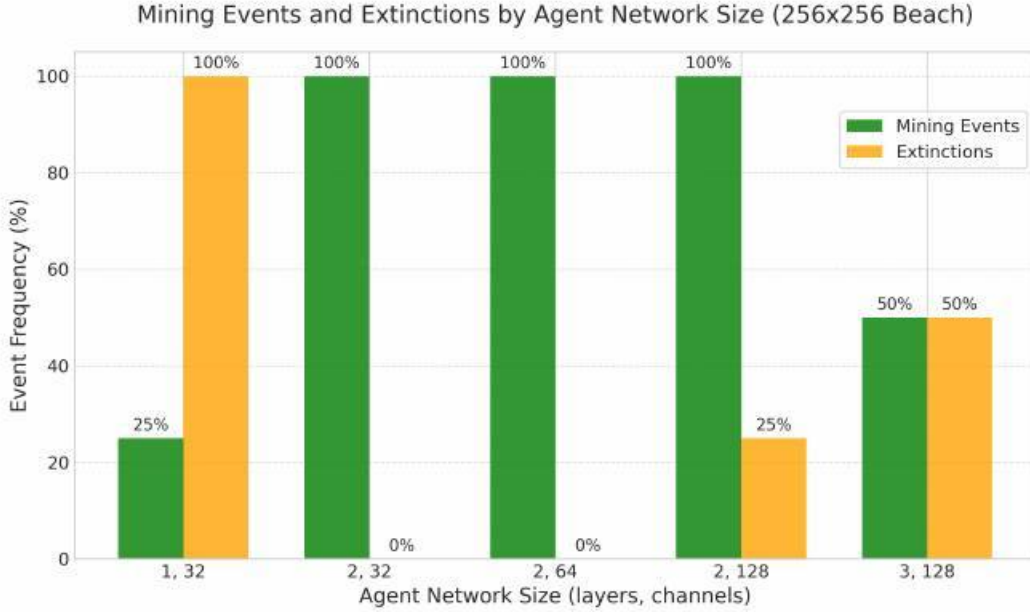


Figure 19: Frequency of mining and extinction events in a 256×256 **Beach** world for different agent policy network sizes. Frequencies are computed across four random seeds for each setting.

F.6 VARYING VISION RANGE

We assess how vision range and viewing geometry shape predation and foraging in the 512×512 **Ocean** world with (RCV+A) agents. We evaluate 3×3 , 7×7 , and 9×9 visual ranges. For each range, we compare the two configurations in Figure 21 a) Around, where agents see cells surrounding

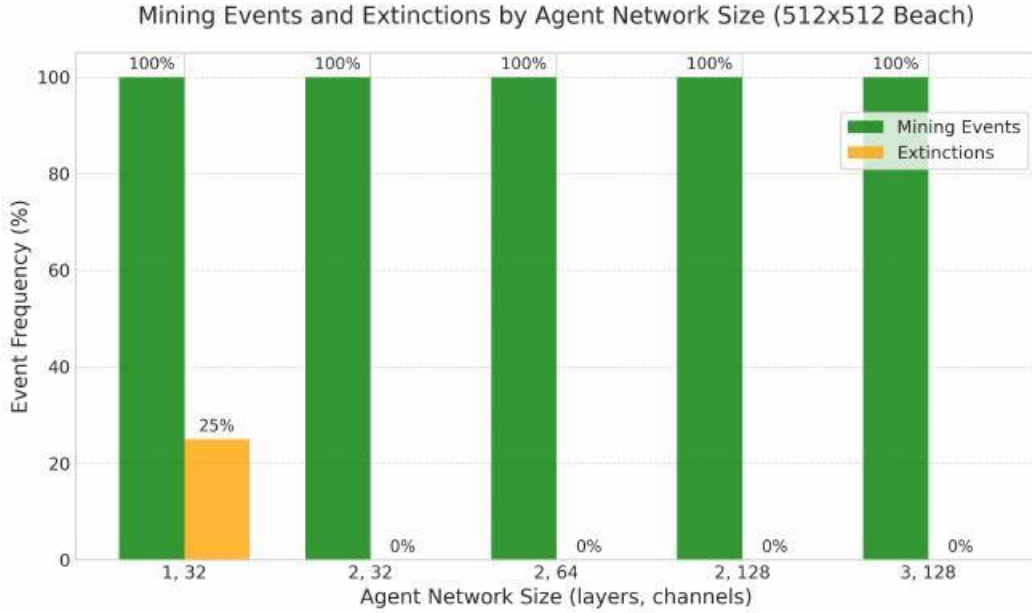


Figure 20: Frequency of mining and extinction events in a 512×512 **Beach** world for different agent policy network sizes. Frequencies are computed across four random seeds for each setting.

them, and b) Front, where agents do not see behind themselves. The two configurations expose the same total number of cells. The main vision experiments in Section 4.2 use the 7×7 Around setting.

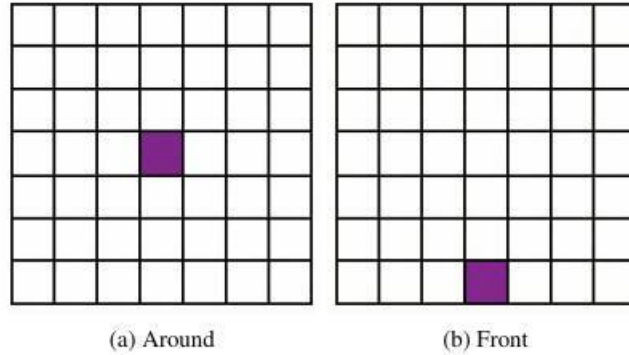


Figure 21: Around and Front viewing configurations for the 7×7 vision range. The purple cell is an agent facing the top of the page.

These results show that 3×3 vision behaves much like no vision at all (Blind), whereas 9×9 vision closely matches 7×7 . Thus, vision has a clear useful range in this environment, followed by diminishing returns. This outcome is consistent with the memoryless policy architecture: peripheral features in a 9×9 window are unlikely to change the optimal immediate action relative to a 7×7 window often enough to provide a strong additional learning signal.

At the 7×7 and 9×9 ranges, the Around configuration is modestly stronger for population stability and attack/eat effectiveness, yielding higher average population , biomass utilization , and homicides per attack than the Front configuration. A plausible explanation is that Around allocates more visible cells to nearby locations, which are more relevant for immediate decisions than the more distant cells exposed by Front. At 3×3 , the pattern reverses: Front performs better than Around, indicating that when the total visual field is extremely small, one additional cell ahead is more valuable than one cell behind, while the Around setting remains closest to Blind behavior.

Table 12: Foraging and predation data in the 512×512 **Ocean** world for vision agents (**RCV+A**) with varying vision ranges and viewing configurations. Refer to Tables 5 and 6 for definitions of the metrics represented by each icon. For each seed, the value for each metric was averaged across all steps starting at step 500k to allow them to stabilize. Reported values are the mean across the four seeds and the standard deviation of the four seed averages.

Vision Range	Viewing Configuration						
3×3	Blind	7212 ± 2496	.181 ± .060	.435 ± .090	.392 ± .033	.032 ± .006	.307 ± .090
	Around	10308 ± 4016	.258 ± .105	.352 ± .014	.427 ± .022	.040 ± .030	.388 ± .195
	Front	13863 ± 520	.351 ± .023	.336 ± .008	.437 ± .009	.026 ± .006	.470 ± .145
7×7	Around	16739 ± 122	.373 ± .003	.259 ± .000	.532 ± .001	.015 ± .000	.751 ± .007
	Front	15255 ± 623	.343 ± .015	.253 ± .001	.540 ± .004	.015 ± .000	.690 ± .041
9×9	Around	16744 ± 117	.372 ± .002	.257 ± .001	.534 ± .001	.015 ± .000	.728 ± .009
	Front	15211 ± 1416	.342 ± .033	.253 ± .001	.541 ± .007	.015 ± .000	.664 ± .050

F.7 VARYING ATTACK STRENGTH

We test how attack strength shapes emergent behavior and population dynamics in the 256×256 **Ocean** world. The sweep uses three settings: 1) Full Attack, which deals 10 HP of damage and instantly kills any agent in the attack radius; 2) Half Attack, which deals 5 HP of damage; and 3) Quarter Attack, which deals 2.5 HP of damage. The main **Ocean** experiments in Section 4.2 use Full Attack.

Reducing attack strength from Full Attack to Half Attack sharply lowers both attack frequency and attack effectiveness: attacks per agent and homicides per attack both drop (Figure 22). Lower attack strength therefore permits larger populations, more movement and eating, and greater biomass utilization. Reducing strength further to Quarter Attack produces diminishing additional reductions in violence and diminishing additional population gains. The mechanism is twofold: weaker attacks require more hits to produce a homicide, and because each attack is less useful, selection favors attacking less strongly. Agents in lower-attack settings therefore evolve less skillful predation and allocate more behavior to foraging.

F.8 VARYING AGENT HP

We investigate whether reducing child starting HP and lowering healing rate creates meaningful pressure for agents to ‘grow’ over their lifetime. The healing-rate parameter determines how much HP an agent regenerates per healing step by converting available energy into HP, so agents can influence healing only by maintaining sufficient energy.

We compare two settings: 1) Full HP, where child HP starts at 5 and healing rate is 1 HP per 0.1 energy, and 2) Low HP, where child HP starts at 1 with healing rate 0.5. Maximum agent HP is 10 in both settings. The main experiments (Section 4) use Full HP.

The two settings produce no substantial difference in homicides per attack, eat-action frequency, or biomass utilization (Figure 23). Full HP supports slightly larger populations, while Low HP produces less frequent attacking and slightly more movement. A plausible explanation is that low-HP newborn agents must prioritize early-life foraging to reach full size, shifting behavior away from attacks and toward movement.

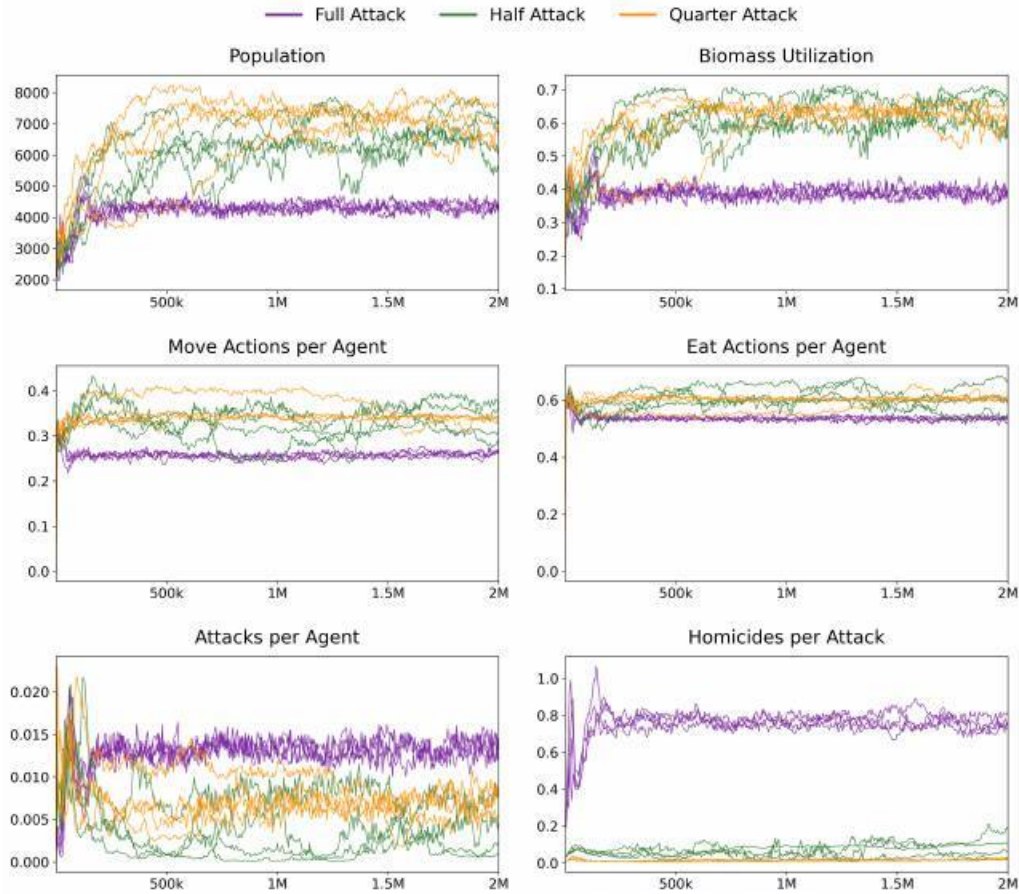


Figure 22: Population and action frequency data over 2M steps in a 256x256 Ocean world for (RCV+A) agents across Full Attack, Half Attack, and Quarter Attack strength settings. Four seeds are plotted per attack strength setting.

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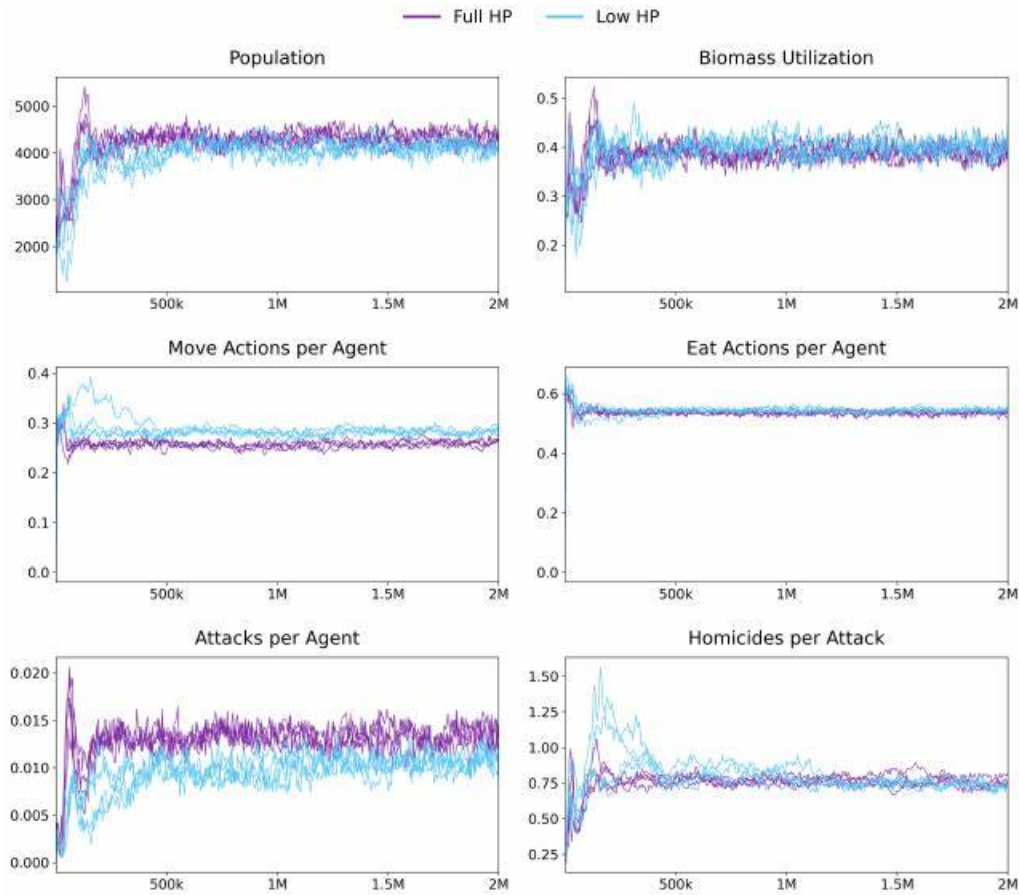


Figure 23: Population and action frequency data over 2M steps in the 256×256 **Ocean** world for (RCV+A) agents across Full HP and Low HP settings. Four seeds are plotted per HP setting.